

Objects as Closures

Constraint Meshing as a Primitive of Mathematical Construction

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ABSTRACT

A mathematical object can be the residue of a relation rather than an independently existing element of either structure that produces it. We develop a constraint-first method in which two incomplete, independently defined constraints are imposed simultaneously and their residuals are permitted to cancel; the resulting zero set — the *meshing* — is the observable. The method is demonstrated on the Bailey–Borwein–Plouffe family. In the base-64, period-6 tower, pole geometry alone leaves a two-dimensional space of admissible coefficient vectors and the zero-sum condition alone leaves a five-dimensional space; neither selects anything. Together they select a unique line, and that line is $9 \cdot [4, -2, -1, -1]$ — the same coefficient signature carried by the base-16 formula for π . The signature is derived rather than observed: it is forced by the single condition $\text{Re}(z_{\text{char}}) = 1$ before either tower is specified. Extending the construction continuously in radix yields a family-indexed transfer geometry, $b = \sec^n \theta$, under which the two towers are angular locks on distinct curves that share an angle but never a radix. Throughout, the object of interest satisfies both constraints while being selected by neither, and vanishes when either is removed. We argue that in constructions of this kind the observable is generated by the joint closure rather than possessed by either constraint, and that a relation-first description is therefore more accurate than an object-first one.

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

1. A constant that is nowhere

Take a three-digit number. Reverse it. Subtract the smaller from the larger. Reverse that difference and add. The answer is 1089. It is always 1089. It does not matter which number you started with.

This is a party trick, and it is also the entire subject of this paper. The interesting question is not why the answer is constant. It is *where the constant lives*. It is not in the number you chose, because any number works. It is not in the subtraction, which by itself produces nine different answers. It is not in the reverse-and-add, which by itself produces nothing at all until it is handed a difference. The 1089 is not stored anywhere in the procedure. It appears at the point where two operations, each of which destroys something the other preserves, are composed.

Watch it happen. Write the number as $100a + 10b + c$ and its reversal as $100c + 10b + a$. Subtract:

$$(100a + 10b + c) - (100c + 10b + a) = 99 \cdot (a - c)$$

The middle digit is gone. Not reduced, not carried — gone, and it never mattered. Reversal is a reflection, and b sits on the mirror's axis; a point on the axis maps to itself, and self minus self is zero. What survives is the span between the two off-axis digits. Three free parameters have become one.

Now the second operation. The difference is $99k$ for some span k between 1 and 9, which as three digits reads $(k-1, 9, 10-k)$. Reverse it and add: the outer digits pair as $(k-1) + (10-k) = 9$ in both the hundreds and units places, and the middle contributes $9 + 9 = 18$. The result is $900 + 180 + 9 = 1089$, and **k has cancelled**. The one surviving parameter is annihilated by the second operation exactly as the first was annihilated by the first.

So 1089 depends on neither free parameter. The first operation kills the middle and keeps the span; the second kills the span. What is left is what neither operation could move. It is not a property of the starting number, nor of either operation taken alone — the subtraction alone has nine possible outputs and the addition alone has no output at all. The constant exists only in the composition, and it is precisely what survives having everything else removed.

This paper is about that structure, made exact and taken seriously. We will show that a coefficient vector governing a well-known family of formulae for π has the same status: it satisfies two constraints, is selected by neither, and ceases to exist when either is withdrawn. We will show this by computation rather than analogy. And we will argue that when a mathematical object behaves this way, the honest description is not that it was discovered somewhere but that it was *produced at a closure*.

2. What the trick was doing

Name the two operations. The first constrains what can be distinguished: the reflection makes the axis unreadable and the span readable, fixing which differences the procedure is able to see. Call this constraint C_0 . The second constrains what change is permitted: the reverse-and-add closes the remaining freedom, forcing the result to be independent of the span. Call this C_1 .

Neither constraint has a distinguished output on its own. C_0 maps 900 starting numbers onto nine values; that is not selection, it is a nine-fold reduction. C_1 has no input until C_0 has run. It is only when both act on the same object that a single value is picked out, and the value picked out is the one that neither constraint is able to vary. Write this as a condition rather than a procedure:

$$\mathcal{M}(C_0, C_1; x) = 0$$

The object of interest is the zero set of the joint condition. The crucial property, and the reason this is not merely a restatement of "solve both equations," is that the joint condition can be satisfied where neither separate condition is decisive. $\mathcal{M}(C_0, C_1) = 0$ does not require, and does not follow from, C_0 or C_1 having a distinguished solution alone. A structure selected this way is a **meshing**, and it has a testable signature: remove either constraint and the selection is destroyed. Not weakened — destroyed. The object does not relocate; the space of candidates simply reopens and there is no longer anything distinguished in it.

2.1 Why simultaneity is not optional

The two constraints must be applied at once, to the same candidate. If they are applied in sequence — satisfy C_0 , fix the result, then check C_1 — whatever C_0 left undetermined has been frozen into the intermediate object, and C_1 can no longer distinguish that residue from genuine structure. The freezing is the error, and it is invisible afterwards because the frozen choice looks like data.

This is the operating principle of a balanced signal line, and the analogy is exact rather than decorative. A balanced line does not carry two independent copies of a signal to be compared. It carries the signal and its inverse on two conductors, and the receiver takes their *difference*. Interference picked up along the route appears on both conductors with the same sign and cancels in the subtraction; the signal appears with opposite signs and survives, doubled. The noise immunity is not a consequence of having two measurements. It is a consequence of subtracting them simultaneously. Take the two readings at different times, or compare them for agreement instead of differencing them, and the immunity is gone.

2.2 Two incomplete readings, one exact result

It is tempting to formalize the two-constraint test as a logical conjunction: accept the candidate if the forward check passes and the return check passes. This is wrong, and the way it is wrong is the substance of the method.

Forward reading	Return reading	Conjunction says	Correct rule
closes	closes	accept	accept
incomplete	closes	reject	reject
closes	incomplete	reject	reject

Forward reading	Return reading	Conjunction says	Correct rule
incomplete	incomplete	reject	accept if residuals cancel

The fourth row is the point. If "incomplete" means that a residual deviation remains, then two readings whose residuals are equal and opposite sum to zero. Each reading is individually wrong. Jointly they are exact. The acceptance condition is not that both pass but that the combined residual closes:

$$r_{forward} + r_{return} = 0$$

A conjunction cannot express this, because a conjunction has no access to the *magnitude* of the failures — only to whether they occurred. Two failures that annihilate are indistinguishable, under conjunction, from two failures that compound. The correct operation is a sum in the residual space, and its zero is the object. In Section 5 this appears concretely: one construction determines the structure of a numerator but not its scale, a second determines a relation but not why its coefficients are what they are, and the scale is fixed only when the two are closed against each other. Neither is complete. The closure is exact.

2.3 The admissibility test

This gives a mechanical criterion for deciding whether a candidate structure is genuinely a meshing or merely a property of one constraint dressed up as a discovery. Remove C_0 and recompute the admissible space; remove C_1 and recompute. If the candidate remains distinguished under either removal, it belonged to that constraint alone and carries no joint information. Only if both removals destroy the selection has anything been found that neither input contained. We will apply this test explicitly, with dimensions counted, rather than asserting it.

3. A harder case: where does π 's formula live?

In 1995 Bailey, Borwein and Plouffe found that π can be written as a series in which each term is a rational function of the index divided by a power of 16, and that this permits extraction of an individual hexadecimal digit of π without computing the preceding ones. The formula's numerator carries a specific set of small integer coefficients. Those coefficients are conventionally regarded as a discovered fact about π — a lucky arrangement that happens to work.

We ask the 1089 question of them. Where do the coefficients live? Are they a property of π , of the base 16, of the period 8? Or are they, like 1089, what remains after two independent constraints have each removed what the other preserves?

To answer this we need a second instance, because a single formula cannot distinguish a property from a residue. We construct one. Let

$$P(t) = b - t^n, \quad S(r) = \sum_{k \geq 0} 1 / (b^k (nk + r)) = b \int_0^1 t^{r-1} / P(t) dt, \quad r = 1 \dots n,$$

so that a digit-extraction formula is a rational combination $\sum_r c_r S(r)$ evaluating to a target constant, and the object under constraint is the coefficient vector $(c_1, \dots, c_n)$. Each $S(r)$ is a rational combination of a fixed finite set of transcendental atoms — logarithms and π -type terms — so the passage from coefficient vectors to atom combinations is a rational linear operator Φ . The two constraints will be stated as conditions on Φ .

The Bailey–Borwein–Plouffe case is $b = 16$, $n = 8$. We will work the case $b = 64$, $n = 6$ in full, because it is new and because having two cases is what makes the argument possible.

4. The hexagonal tower, and the vanishing

Take $b = 64$, $n = 6$. Over the rationals,

$$64 - t^6 = -(t - 2)(t + 2)(t^2 - 2t + 4)(t^2 + 2t + 4).$$

The six roots are the sixth roots of 64, lying on the circle of radius 2 at $0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ$. The root modulus $\rho = 2$ satisfies $\rho^6 = 64 = b$. Computing the six shell integrals and reducing to the transcendental basis $\{\log 2, \log 3, \log 7, \sqrt{3} \cdot \pi, \sqrt{3} \cdot \arctan(\sqrt{3}/2)\}$ gives a 6×5 matrix for Φ . Its rank is 5, so Φ is surjective onto the atom space and its kernel is one-dimensional:

$$\ker \Phi = \text{span}\{z\}, \quad z = (16, -24, -8, -6, 1, 0), \quad \sum z = -21.$$

Now state the two constraints as conditions on a coefficient vector c .

- **C_0 (pole geometry).** The formula must read purely on the $\sqrt{3} \cdot \pi$ atom: the projections of $\Phi(c)$ onto the four non- π atoms must vanish. Geometrically, the numerator must annihilate the poles that do not sit at the characteristic angle.
- **C_1 (zero-sum, or DC-blocking).** The coefficients must sum to zero: $\sum_r c_r = 0$. The instrument must have no response to a constant.

These are independently motivated. The first is about which poles survive; the second is about the numerator's behaviour at $t = 1$. Neither refers to the other, and neither mentions any target vector.

4.1 Neither constraint selects anything

Impose C_0 alone. The admissible space drops from six dimensions to two. That is a substantial reduction and it selects nothing: a two-dimensional space contains infinitely many directions, and no member of it is distinguished.

Impose C_1 alone. The admissible space drops from six dimensions to five. Again, nothing is selected.

Impose both, simultaneously, on the same vector. The admissible space is **one-dimensional** — a unique line. Scaled to integers it is

$$(36, -18, -9, -9, 0, 0) = 9 \cdot (4, -2, -1, -1, 0, 0).$$

The active weights are $9 \cdot [4, -2, -1, -1]$. This is the same coefficient signature that the base-16 formula for π carries. It was not supplied to the computation; the computation was given two constraints and no target.

4.2 The vanishing, counted

The dimensional bookkeeping is the argument, so it is worth setting down plainly.

Constraints imposed	Admissible dimension	Is anything selected?
none	6	no
C_0 only (pole geometry)	2	no
C_1 only (zero-sum)	5	no
C_0 and C_1 together	1	yes — a unique line

Two further facts sharpen this. First, the vector $(4, -2, -1, -1, 0, 0)$ *satisfies* C_0 , and it *satisfies* C_1 — it lies inside both admissible spaces. It is selected by neither, because each of those spaces has dimension greater than one. Membership is not selection. Second, the constraints are independent in the strict sense: C_0 removes four dimensions, C_1 removes one, and together they remove five, exactly the sum. There is no overlap. Neither constraint is doing part of the other's work.

So the coefficient vector satisfies both constraints, is selected by neither, and is selected by their intersection. Withdraw C_0 and it is one direction among infinitely many in a two-dimensional space, with nothing to distinguish it. Withdraw C_1 and it is one among infinitely many in a five-dimensional space. It does not move elsewhere. There is simply no longer anything there to find. This is the 1089 structure, in six dimensions, with the vanishing measured rather than described.

5. Two incomplete constructions, one closed ledger

Each tower, taken by itself, cannot finish. Writing the numerator in factored form makes the incompleteness visible:

$$N(t) = c \cdot (1 - t) \cdot A(t)$$

Three factors, three sources. $A(t)$, the pole annihilator, is fixed by which roots must be cancelled — that is C_0 . The factor $(1 - t)$ is fixed by the zero-sum condition — that is C_1 . The scalar c is fixed by neither. It is the residual: the single-tower construction determines the numerator's entire structure and leaves its scale open.

The base-16 tower has $c = 1$. The base-64 tower has $c = 9/4$. Neither tower can say why. The scale is determined only by closing the two towers against one another, and when that is done the ledger closes exactly:

$$\sqrt{3} = (64/16) \cdot (9/4) \cdot (\sqrt{3}/9)$$

The three factors come from three independent places. The ratio $64/16 = 4$ is fixed by the bases, which are themselves forced (Section 6). The ratio $\sqrt{3}/9$ is the ratio of the two bare integrals, fixed by pole geometry. The ratio $9/4$ is the scale — the residual. The identity holds if and only if the scale ratio equals $\sqrt{3} \div (4 \cdot \sqrt{3}/9) = 9/4$, which is exactly what it is. The scale was never free. It is the unique value that makes the cross-construction residual vanish.

This is the fourth row of the table in Section 2.2, realized. The forward construction is incomplete: it has an undetermined scale. The return relation is incomplete: it exhibits a coefficient match it cannot explain. Neither is a proof. Their residuals cancel, and the result is exact.

6. The signature is derived before either tower exists

A shared coefficient vector across two constructions invites the objection that the sharing is coincidence, or that the second construction was built to match the first. The objection is answered by deriving the signature from a condition that mentions neither tower.

The condition is that the real part of the surviving characteristic pole equals one:

$$\text{Re}(z_{\text{char}}) = 1$$

This is the condition under which the surviving pole contributes to π through an arctangent-type integral whose argument structure requires unit real part. Everything else follows by elementary algebra.

The base is forced. $\text{Re}(z_{\text{char}}) = 1$ means $p \cdot \cos\theta = 1$, hence $\text{base} = (1/\cos\theta)^n = \sec^n\theta$. For $n = 6$, $\theta = 60^\circ$ this gives $2^6 = 64$. For $n = 8$, $\theta = 45^\circ$ it gives $(\sqrt{2})^8 = 16$. The bases were not chosen; they are outputs.

The annihilator's sub-leading coefficient is forced to 2. The n roots of the denominator sum to zero. The characteristic pair contributes $2 \cdot \text{Re}(z_{\text{char}}) = 2$. The real roots contribute zero, occurring in $\pm$ pairs. Hence the annihilated roots sum to -2 in both towers.

The annihilator's constant term is forced to 4. Each annihilated conjugate pair contributes $|\text{root}|^2 = \rho^2$, and $\rho^{2d} = \text{base}^{(2d/n)}$ equals 4 in both towers, since $64^{(1/3)} = 16^{(1/2)} = 4$.

So every annihilator in this class has top coefficients $[1, 2, 4]$. Applying the zero-sum multiplier $(1 - t)$ takes successive differences of those coefficients, and the non-zero pattern that results is

$$[4, -2, -1, -1]$$

No step of this is observation. The signature is the coefficient fingerprint of $\text{Re}(z_{\text{char}}) = 1$, and it is fixed before a base, a period, or a target constant has been named. What the two towers share is not a lucky vector but membership in the condition that produces it.

7. The constant is a lock on a curve

So far the radix has been an integer chosen in advance. Releasing it turns a pair of isolated formulae into a geometry. Write the two source constraints of the frame-winding construction in their continuous form: C_0 requires the endpoint to be the frame mark, $u = \cos\theta$; C_1 requires the endpoint raised to the shell spacing to be the reciprocal radix, $u^n = 1/b$. Eliminating u gives the joint zero set,

$$b = \sec^n \theta$$

equivalently $\theta = \arccos(b^{-1/n})$. This is not a fitted law; it is the elimination of a variable between the two constraints. For fixed n it is a single smooth curve in the (b, θ) plane, and the base-64 tower sits on it at $b = 64$, $u = 1/2$, $\theta = 60^\circ$ with closure residual of order 10^{-16} . Perturbing b from 64 to 65 moves the angle continuously to 60.085332° , and reconstructing b from that angle returns to the manifold with residual of order 10^{-18} . The lock is a point on a continuum, not an isolated coincidence.

The two towers, however, do not lie on the same curve. Setting $\theta_6(b) = \theta_8(b)$ requires $b^{-1/6} = b^{-1/8}$, that is $b^{-1/24} = 1$, whose only solution is $b = 1$. For every $b > 1$ the $n = 6$ and $n = 8$ curves are strictly distinct and never meet. What the two towers share is not a radix but an angle:

Characteristic angle θ	b on the $n = 6$ curve	b on the $n = 8$ curve
45°	$\sec^6(45^\circ) = 8$	$\sec^8(45^\circ) = 16$
60°	$\sec^6(60^\circ) = 64$	$\sec^8(60^\circ) = 256$

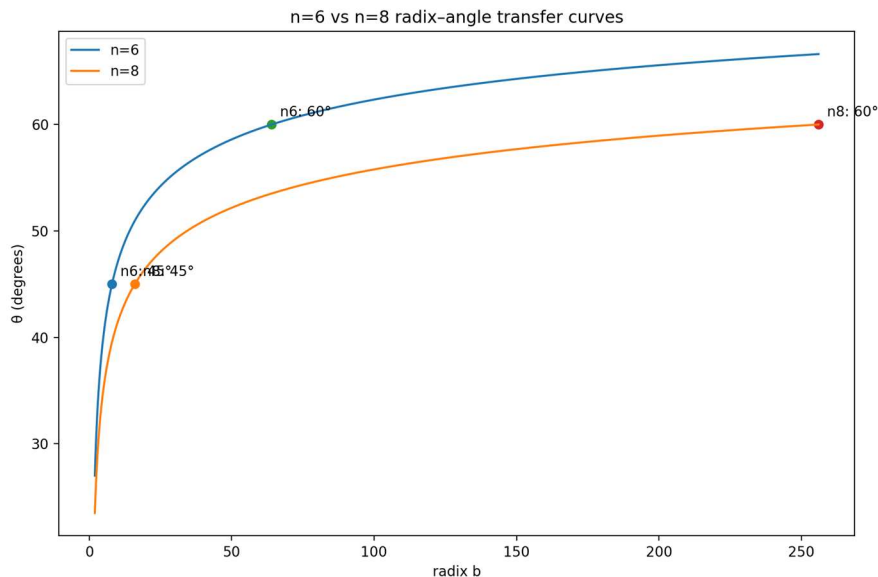


Figure 1. The $n = 6$ and $n = 8$ transfer curves $\vartheta = \arccos(b^{(-1/n)})$. They share the characteristic angles 45° and 60° but reach them at different radices, and they do not intersect for $b > 1$. Both are monotone and concave: the gearing $d\vartheta/db$ becomes finer as the radix grows.

The structure is therefore a family of curves indexed by n , sharing one angular coordinate, with the radix forced by $b = \sec^n \theta$ once the family and the angle are set. The two characteristic angles are distinguished by a clean-radix condition: among candidate angles only 45° and 60° make $\sec^n \theta$ an integer power of two for these n , while $\sec^6(30^\circ)$, $\sec^6(36^\circ)$ and $\sec^6(72^\circ)$ are not integers at all.

This changes what a digit-extraction formula is. It is not an isolated identity that happens to hold. It is a **lock** — a point where a continuous constraint-defined curve meets an integrality condition. The formula for π in base 16 is one such lock; the formula for $\sqrt{3}\pi$ in base 64 is another; and the curve running between and beyond them is populated everywhere by admissible non-integer configurations that are not formulae only because their radices are not clean. The constant does not sit at a special place because it is special. It sits where the curve crosses a condition.

8. What the construction shows

Three things are established, each by executed computation and none by analogy.

The coefficient vector $g[4, -2, -1, -1]$ is selected by the intersection of two independent constraints and by neither of them alone. The dimensions are $6 \rightarrow 2$ under pole geometry, $6 \rightarrow 5$ under the zero-sum condition, $6 \rightarrow 1$ under both, with the reductions adding exactly and no overlap between them. The vector belongs to both admissible spaces and is distinguished in neither.

The signature $[4, -2, -1, -1]$ is a consequence of $\text{Re}(z_{\text{char}}) = 1$, derivable before any tower is specified, and it forces the bases 64 and 16 rather than being fitted to them. The base-64 tower is therefore a genuine second member of the class and not a construction reverse-engineered to match the first.

The two towers lie on distinct curves of the family $b = \sec^n \theta$, sharing characteristic angles but never a radix, and each formula is a lock where its curve meets an integrality condition on the radix.

8.1 The interpretation, and its limits

The coefficient vector was not found inside either constraint. It appeared only where the constraints closed. That sentence is a statement about a computation, and it is the strongest thing the computation supports. The question worth asking is how far it generalizes.

If a mathematical structure can be produced at the intersection of two constraints while being possessed by neither, then in constructions of that kind the object is not the primitive. The relation is. What we ordinarily call the object is the readable residue of a relation that has closed — a lock on a curve, an intersection of independently swept conditions, a value that neither input can move. Under this description a constant is not a thing that exists and is then discovered; it is what appears when two constraints are made to hold at once, and it stops appearing when either is released.

We do not claim to have proved that mathematical objects in general lack independent existence, and the reader should resist that reading. The BBP family is one construction. What it establishes is that *at least here*, the object-first description is not merely inconvenient but inaccurate: it attributes to a coefficient vector a residence it demonstrably does not have. Whether the same holds elsewhere is a research question, and it is answerable in the same way — by counting dimensions before and after removing a channel, and seeing whether the structure survives.

What this suggests is a change of default. Where an object-first description asks *what is this constant and where does it come from*, a relation-first description asks *which two constraints close here, and what do they leave*. The second question has an answer in this paper. The first, for the BBP coefficients, has never had one.

9. What remains open

The family relation $b = \sec^n \theta$ has three coordinates, and only two are derived. The angle is the shared coordinate; the radix is forced once family and angle are set. The family index n is not derived. The underlying classification admits a candidate set, and the two built towers are $n = 6$ and $n = 8$, but nothing here explains why those and not the others. That is the sharpest open problem the construction produces, and it is well posed: identify the constraint whose closure selects n .

Two further questions follow from it. It is not settled whether the signature $[4, -2, -1, -1]$ is forced for every member of the class or only for those on the clean-radix locus; the derivation in Section 6 uses the condition $\text{Re}(z_{\text{char}}) = 1$, and the relationship between that condition and the integrality of $\sec^n \theta$ is not yet established. And the transfer geometry measures the gearing without explaining why the clean-radix condition falls where it does.

Each of these is a constraint not yet closed, and the method for closing them is the one this paper has used: state the two conditions, impose them simultaneously, and read what neither can move.

$$\mathcal{M}(C_0, C_1; \mathbf{x}) = 0$$

Computational note. Every numerical statement in this paper was produced by executed code before being written. The factorization of $64 - t^6$; the rank-5 operator and its kernel $z = (16, -24, -8, -6, 1, 0)$ with $\Sigma z = -21$; the admissible dimensions 6, 2, 5, 1 and the independence of the two constraints; the selected line $g \cdot [4, -2, -1, -1]$; the ledger identity $\sqrt{3} = (64/16)(9/4)(\sqrt{3}/9)$; the base-forcing $\sec^6(60^\circ) = 64$ and $\sec^8(45^\circ) = 16$; the non-intersection condition $b^{(-1/24)} = 1$; and the perturbation $64 \rightarrow 65$ with closure residuals of order 10^{-16} and 10^{-18} . Computation in Python with *mpmath*, *sympy* and *numpy* at 40–50 decimal digits where precision is material. Source for the constraint compiler and the transfer engine accompanies this paper.